

The Desert Thermal Penalty on an Air-Cooled Heat-Pipe Microreactor

Phase 1 Results Summary — Derating Curve for Saudi Siting

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Abstract

An eVinci-class $15\text{ MW}_{\text{th}}/5\text{ MW}_{\text{e}}$ heat-pipe microreactor rejects no water: its power is produced by a *recuperated open-air Brayton* cycle — essentially a ground-based jet engine whose compressor breathes ambient air, and which recycles its own exhaust heat to save fuel. We quantify how Saudi ambient heat (25°C to 55°C) derates the net electric output, using a calibrated cycle model refined in three stages and validated by CFD. The net penalty at 55°C is bracketed by a control-strategy band: -17.7% (conservative baseline) to -8.7% (optimistic, CFD-validated). A dedicated OpenFOAM solve of the finned-tube heater confirmed the air-side heat-transfer law ($Nu = 45.9$, within 15% of Briggs–Young) and showed the penalty is *robust* to that law. The desert derating is dominated by compressor-inlet thermodynamics, giving a physically defensible per-site input for the Phase-2 siting study.

1 Objective and reference system

An *open-air* Brayton draws its working fluid from the atmosphere and exhausts it, so the compressor-inlet temperature equals the ambient temperature. This is the classic gas-turbine “hot-day derating” (the same reason cars and aircraft lose power on a hot day or at altitude, where the air is thinner), and it makes the water-free design that suits arid siting also the design most sensitive to desert heat. The reactor core delivers a fixed 15 MW_{th} through sodium heat pipes regardless of weather; all ambient sensitivity lives in the power conversion system. Reactor-scale facts are taken from the IAEA ARIS 2024 datasheet (Table 1); the finned-tube heat-exchanger geometry is not public and was specified as a representative standard bank.

Table 1: Reference design point (calibration v1.1).

Thermal / electric	15/5 MW	Pressure ratio	3.0
Power conversion	recup. open-air Brayton	Compressor / turbine η	0.82 / 0.85
Turbine-inlet temp. (solved)	742°C	Recuperator ε	0.88
Net efficiency @ 25°C	33%	Mech./parasitic η_{mech}	0.93

2 Methodology

The cycle output is $P_{\text{net}} = \eta_{\text{mech}} \dot{m} (w_t - w_c)$, where the compressor and turbine specific works follow the isentropic-efficiency relations with $w_c \propto T_1$ (inlet temperature). The curve was built in three drafts, each removing one assumption:

- v1** Air-standard recuperated Brayton, calibrated by fixing η_{mech} and *solving* the turbine-inlet temperature to hit 33% net at 25 °C (lands in the Na heat-pipe band). Firing temperature held fixed $\Rightarrow$ conservative baseline.
- v1.5** Replaces v1's *assumed* recuperator effectiveness with sized ε -NTU heat exchangers ($\varepsilon = \text{NTU}/(1+\text{NTU})$ balanced recuperator, $UA \approx 401 \text{ kW/K}$; $\varepsilon = 1 - e^{-\text{NTU}}$ isothermal condensing-Na heater, $UA \approx 152 \text{ kW/K}$) coupled to the reactor energy balance, with turbine-inlet temperature allowed to float up (optimistic bound). One guess remained: the air-side law $UA \propto \dot{m}^{0.6}$.
- v2** Replaces that last guess with a steady compressible OpenFOAM solve (`rhoSimpleFoam`, $k-\omega$ SST, 3.29 M cells, $y_{\text{mean}}^+ = 0.24$) of a finned-tube heater unit cell at the design Reynolds number. Convection is reduced by the Buckingham- π theorem to $Nu = f(Re, Pr)$, so one validated unit cell represents the whole bank.

3 Results

The derating band. Figure 1 is the headline. The conservative baseline (v1) loses -17.7% of net power at 55 °C ($\approx 0.60 \text{ \%/}^\circ\text{C}$ over 25 °C to 45 °C); the optimistic, CFD-validated bound (v2) loses -8.7% ($\approx 0.15 \text{ \%/}^\circ\text{C}$). The true output lies within this control-strategy band, and the conservative edge is carried into the siting study. Even the conservative slope is about $4\times$ the Carnot-only ceiling drop ($0.14 \text{ \%/}^\circ\text{C}$): the difference is the real-cycle amplification. At 55 °C the loss factorizes as 0.863 (rising compressor work) $\times 0.953$ (falling intake mass flow) $= 0.823$, i.e. compressor-inlet thermodynamics dominate.

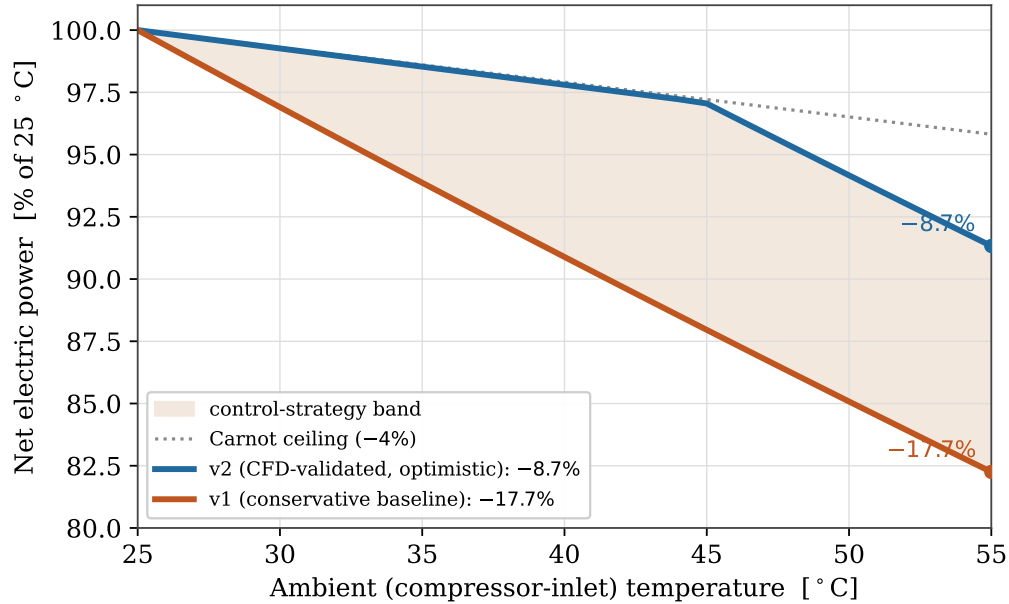


Figure 1: Net electric power vs. ambient temperature. The band spans the conservative fixed-firing baseline (v1, -17.7%) and the optimistic CFD-validated bound (v2, -8.7%); the dotted line is the unavoidable Carnot-ceiling drop (-4%). The gap between the real curves and Carnot is the real-cycle amplification of the second law.

CFD validation. The heater solve yields a wall heat rate of 256.3 W ; reduced through the log-mean temperature difference (the proper average wall-to-air temperature gap) it gives $h \approx 110 \text{ W/m}^2\text{K}$ and a Nusselt number $Nu = 45.9$ (a cooling score — the moving air pulls heat off the tube about 46 times

better than a still layer would). Against the Briggs–Young correlation at $Re = 8368$ this is -15.2% — within the $\pm 20\%$ acceptance band, so the air-side heat-transfer model is **validated** (Fig. 2, Table 2). Friction did not validate ($+489\%$, a measurement-domain mismatch) and is therefore reported but not used. Injecting the validated law ($UA \propto \dot{m}^{0.454}$) into v1.5 moves the penalty by only $+0.08$ points ($-8.8\% \rightarrow -8.7\%$): because the ambient-driven mass flow varies only $\sim 5\%$, the exact heater law is second order. The CFD’s role was to *confirm* that, and it did.

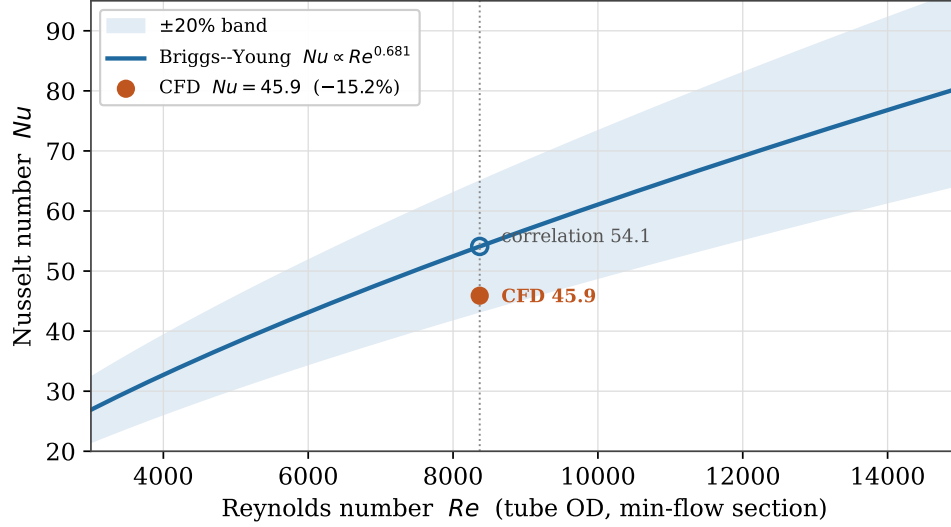


Figure 2: Single-point CFD validation of the air-side Nusselt number against the Briggs–Young correlation. The CFD point ($Nu = 45.9$) sits inside the $\pm 20\%$ band, 15.2% below the correlation — close agreement for a first-principles solve, certifying the $Nu(Re)$ law used across the operating range.

Table 2: Results summary. Each model draft removes one assumption; the penalty at 55°C converges.

Model	Refinement over previous	Penalty @ 55°C	Slope (25°C to 45°C)
v1	air-standard cycle, fixed firing temp. (conservative)	-17.7%	$0.60\%/^\circ\text{C}$
v1.5	ε -NTU sized heat exchangers, floating TIT	-8.8%	$0.15\%/^\circ\text{C}$
v2	+ CFD-validated air-side law $UA \propto \dot{m}^{0.454}$	-8.7%	$0.15\%/^\circ\text{C}$
CFD validation: $Nu = 45.9$ vs. 54.1 (Briggs–Young) = -15.2% [validated] ; friction $+489\%$ [not used]			

4 Conclusion

The desert thermal penalty for an eVinci-class open-air Brayton microreactor is real and large: net electric output falls by 9% to 18% between 25 and 55°C , several times the Carnot-only limit, driven by compressor-inlet thermodynamics rather than the heat-rejection hardware. A validated OpenFOAM air-side setup confirmed the heat-exchanger law and showed the result is insensitive to it. The deliverable is a quantitatively defensible, per-site derating band — the physics-grounded interface that replaces the heuristic “climate” criterion in the Phase-2 siting platform.

5 Key terms in plain language

Term	In plain language, with an everyday example
Open-air Brayton cycle	A ground-based jet engine: it draws in outside air, squeezes it, heats it, and lets it spin a turbine, then blows the air back out. “Open” means it breathes the surrounding air instead of recycling a sealed gas.
Recuperator	A heat-recycler: the turbine’s hot exhaust pre-warms the incoming air before the reactor, so less reactor heat is needed — like exhaling into a scarf so your next breath in is already warmer.
Compressor-inlet temperature	The temperature of the engine’s “first breath”. In the desert this is the hot outside air, and hot air is thinner — the same reason cars and aircraft lose power on hot days or at altitude.
Firing (turbine-inlet) temperature	How hot the gas is when it reaches the turbine blades: hotter gives more power, but it is capped by what the metal and heat pipes can survive — like an oven’s maximum setting.
Mass flow ($\dot{m}$)	Kilograms of air passing through each second, like litres-per-second through a pipe. Hot, thin air packs fewer kilograms per second, so less power.
Carnot ceiling	The best efficiency physics allows between a hot and a cold temperature — a hard limit no engine can beat. A hotter “cold side” (desert air) lowers the ceiling.
Effectiveness (ε)	How close a heat exchanger gets to moving the most heat it possibly could; $\varepsilon = 0.9$ means it captures 90 % of that maximum.
NTU	A heat exchanger’s “size” relative to the flow it must heat — a bigger car radiator has a higher NTU and moves more heat.
CFD	A computer experiment: the air space is chopped into millions of tiny cells and the physics solved in each — a very detailed “wind tunnel on a computer”.
Nusselt number (Nu)	A cooling score — how much better moving air removes heat than still air. $Nu = 46$ means the airflow removes heat about 46 times better than a stagnant layer.
Reynolds number (Re)	Whether the flow is smooth or turbulent: low is smooth like slow honey, high is churning like river rapids. Turbulent flow grabs heat better.
Buckingham- π theorem	A maths shortcut that collapses many variables into a few ratios, so one small test stands in for many cases — like a recipe that depends only on ingredient <i>ratios</i> and so scales to any batch size.
Log-mean temp. difference	The correct average temperature gap between the hot wall and the warming air — a special average, because the gap shrinks as the air heats up along the tube.
Briggs–Young correlation	The “textbook answer”: a published formula from decades of lab experiments on finned tubes, used here to check the simulation.

Sources: `cycle_model/derating_v1.py`, `hx_entu_v1.5.py`, `cf4/` (OpenFOAM ESI 2406); reference facts from IAEA ARIS 2024. Full rationale in `DECISIONS_LOG.md` (D10–D12) and `spec/A1_reference_spec_sheet.md`.